

AKDENİZ UNIVERSITY · Department of Computer Engineering
CSE 413 — Optimization Theory and Applications · Midterm Examination

ANSWER KEY

EXAMPLE PAPER. Written for self-study. It was not issued by the instructor and is not a real past exam.

Question 1 — Modeling

10 points

(a) Model (6 pts: variables 1, objective 2, constraints 3)

x_1 = desks produced per week, x_2 = bookshelves produced per week.

$$\max \quad z = 40x_1 + 30x_2 \quad (\text{profit, } \times 100 \text{ TL})$$

$$\text{s.t.} \quad 2x_1 + x_2 \leq 100 \quad (\text{CNC hours})$$

$$x_1 + x_2 \leq 80 \quad (\text{assembly hours})$$

$$x_1 \leq 40 \quad (\text{desk demand})$$

$$x_1, x_2 \geq 0$$

(b) Classification (4 pts)

Linear: the objective and every constraint are linear in x , so this is an LP.

Continuous if fractional production is allowed (e.g. weekly averages). If whole units are required it becomes an integer LP. Here the LP optimum happens to be integral anyway.

Convex feasible set: each constraint defines a half-space, and an intersection of half-spaces (a polyhedron) is convex.

Question 2 — The Simplex Method

20 points

(a) Standard form (4 pts)

$$\max \quad z = 40x_1 + 30x_2$$

$$\text{s.t.} \quad 2x_1 + x_2 + s_1 = 100$$

$$x_1 + x_2 + s_2 = 80$$

$$x_1 + s_3 = 40$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Initial BFS (origin): $x_1 = x_2 = 0$, $s = (100, 80, 40)$, $z = 0$.

(b) Tableau simplex (12 pts: 4 per correct pivot, including the ratio test)

Tableau 0. Entering x_1 (most negative -40). Ratios $100/2 = 50$, $80/1 = 80$, $40/1 = 40$, so s_3 leaves.

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
s_1	2	1	1	0	0	100	50
s_2	1	1	0	1	0	80	80
s_3	1	0	0	0	1	40	40 ←
z	-40	-30	0	0	0	0	

Tableau 1. Entering x_2 (-30). Ratios $20/1 = 20$, $40/1 = 40$, and the x_1 row has no ratio (coefficient 0), so s_1 leaves.

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
s_1	0	1	1	0	-2	20	20 ←
s_2	0	1	0	1	-1	40	40
x_1	1	0	0	0	1	40	—
z	0	-30	0	0	40	1600	

Tableau 2. Entering s_3 (-20). Ratios: x_2 row skipped (coefficient $-2 < 0$), $20/1 = 20$, $40/1 = 40$, so s_2 leaves.

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
x_2	0	1	1	0	-2	20	—
s_2	0	0	-1	1	1	20	20 ←
x_1	1	0	0	0	1	40	40
z	0	0	30	0	-20	2200	

Tableau 3. Every reduced cost is ≥ 0 , so this tableau is optimal.

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_2	0	1	-1	2	0	60
s_3	0	0	-1	1	1	20
x_1	1	0	1	-1	0	20
z	0	0	10	20	0	2600

(c) Reading the solution (4 pts)

$\mathbf{x}^* = (20, 60)$, $\mathbf{z}^* = 2600$ (i.e. 260,000 TL per week).

The demand constraint $x_1 \leq 40$ is **not binding**: $s_3 = 20$. The CNC and assembly constraints are both tight.

Vertex check: $(0,0) \rightarrow 0 \cdot (40,0) \rightarrow 1600 \cdot (40,20) \rightarrow 2200 \cdot (20,60) \rightarrow 2600 \cdot (0,80) \rightarrow 2400$. The simplex path went $(0,0) \rightarrow (40,0) \rightarrow (40,20) \rightarrow (20,60)$.

Question 3 — Duality and Sensitivity

15 points

(a) Dual (5 pts)

$$\begin{aligned}
 \min \quad & w = 100y_1 + 80y_2 + 40y_3 \\
 \text{s.t.} \quad & 2y_1 + y_2 + y_3 \geq 40 && (x_1 \text{ column}) \\
 & y_1 + y_2 \geq 30 && (x_2 \text{ column}) \\
 & y_1, y_2, y_3 \geq 0
 \end{aligned}$$

(b) Complementary slackness (6 pts)

The demand constraint is slack at the optimum ($s_3 = 20 > 0$), so $y_3 = 0$.

Both $x_1 = 20 > 0$ and $x_2 = 60 > 0$, so both dual constraints must be tight:

$$2y_1 + y_2 = 40, y_1 + y_2 = 30 \Rightarrow y_1 = 10, y_2 = 20$$

$\mathbf{y}^* = (10, 20, 0)$. **Strong duality**: $\mathbf{w}^* = 100 \cdot 10 + 80 \cdot 20 + 40 \cdot 0 = 2600 = \mathbf{z}^*$ ✓

Shortcut: the same numbers appear under s_1, s_2, s_3 in the z -row of the final tableau.

(c) Shadow prices (4 pts)

CNC offer: reject. One extra CNC hour is worth $y_1^* = 10$, which is less than its price of 15, so each hour would lose 5.

Assembly offer: accept. One extra assembly hour is worth $y_2^* = 20$, which is more than its price of 15, so each hour gains 5. This holds only while the current basis stays optimal.

Question 4 — Mathematical Foundations

15 points

(a) Norms (3 pts)

$$\|v\|_1 = 1 + 2 + 2 = 5, \|v\|_2 = \sqrt{1 + 4 + 4} = 3, \|v\|_\infty = 2.$$

The ℓ_1 unit ball is a diamond (a square rotated 45°) with vertices $(\pm 1, 0)$, $(0, \pm 1)$.

(b) Gradient and Hessian (4 pts)

$$\nabla f(x, y) = (2x - 2y + e^x, -2x + 4y), \nabla^2 f(x, y) = [[2 + e^x, -2], [-2, 4]]$$

$$\text{At } (0, 0): \nabla f = (1, 0), \nabla^2 f = [[3, -2], [-2, 4]], f(0, 0) = 1.$$

(c) Strict convexity (4 pts)

Second-order condition: f is strictly convex if $\nabla^2 f(x) \succ 0$ for every x . Check the leading principal minors (Sylvester):

$$D_1 = 2 + e^x > 0, D_2 = 4(2 + e^x) - 4 = 4 + 4e^x > 0 \text{ for all } x.$$

So $\nabla^2 f \succ 0$ on all of $\mathbb{R}^2$, and f is strictly convex.

Alternative proof: $x^2 - 2xy + 2y^2 = (x - y)^2 + y^2$, a sum of convex functions, and e^x is strictly convex.

(d) Taylor model (4 pts)

$$m(p) = f(0) + \nabla f(0)^T p + \frac{1}{2} p^T \nabla^2 f(0) p$$

$$m(p) = 1 + p_1 + \frac{1}{2}(3p_1^2 - 4p_1p_2 + 4p_2^2) = 1 + p_1 + 1.5p_1^2 - 2p_1p_2 + 2p_2^2$$

Question 5 — Gradient Descent

15 points

(a) (2 pts)

$$Q = \text{diag}(2, 10), \text{ eigenvalues } \lambda = 2, 10, \kappa = 10/2 = 5.$$

(b) (3 pts)

$$\nabla f(x) = (2x_1, 10x_2), \text{ so } \nabla f(5, 1) = (10, 10) \text{ and } f(x^{(0)}) = 25 + 5 = 30.$$

$$x^{(1)} = (5, 1) - 0.1 \cdot (10, 10) = (4, 0), f(x^{(1)}) = 16.$$

(c) (3 pts)

Each coordinate updates independently: $x_i \leftarrow (1 - \alpha\lambda_i) x_i$. The method converges iff $|1 - \alpha\lambda_i| < 1$ for every i , which gives $0 < \alpha < 2/\lambda_{\max} = 0.2$.

With $\alpha = 0.25$: the x_1 factor is $1 - 0.5 = 0.5$, so that coordinate shrinks. The x_2 factor is $1 - 2.5 = -1.5$, so x_2 flips sign and grows by $1.5\times$ every iteration. The iteration **diverges**.

(d) Backtracking (5 pts)

Armijo condition: $f(x + \alpha d) \leq f(x) + \alpha \nabla f(x)^T d$. Here $\nabla f^T d = -\|\nabla f\|^2 = -200$, so the test is $f(x + \alpha d) \leq 30 - 20\alpha$.

α	$\mathbf{x} + \alpha \mathbf{d}$	$f(\mathbf{x} + \alpha \mathbf{d})$	$30 - 20\alpha$	Result
1	$(-5, -9)$	430	10	reject
0.5	$(0, -4)$	80	20	reject
0.25	$(2.5, -1.5)$	17.5	25	accept

Accepted $\alpha = 0.25$, new point $(2.5, -1.5)$, $f = 17.5$.

Note: $0.25 > 2/\lambda_{\max}$. Armijo only needs sufficient decrease in this one step, so it can accept a step that would diverge if used as a constant step size.

(e) (2 pts)

$\alpha = 2/12 = 1/6$. The factors are $1 - 2/6 = 2/3$ and $1 - 10/6 = -2/3$.

Worst case $\|\mathbf{x}^{(k+1)} - \mathbf{x}^*\| \leq (2/3)\|\mathbf{x}^{(k)} - \mathbf{x}^*\|$, and $2/3 = (\kappa - 1)/(\kappa + 1)$ with $\kappa = 5$.

Question 6 — Newton and Quasi-Newton Methods

10 points

(a) Newton step (6 pts)

Solve $\nabla^2 f \Delta \mathbf{x} = -\nabla f$ with $\nabla^2 f = \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix}$ and $\nabla f = (1, 0)$.

$\det = 12 - 4 = 8$, $(\nabla^2 f)^{-1} = (1/8) \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$.

$\Delta \mathbf{x}_{\text{nt}} = -(1/8)(4, 2) = (-1/2, -1/4)$.

Check: $3(-1/2) - 2(-1/4) = -1 \checkmark$, $-2(-1/2) + 4(-1/4) = 0 \checkmark$.

Newton decrement: $\lambda^2 = \nabla f^T (\nabla^2 f)^{-1} \nabla f = 4/8 = 1/2$, so **$\lambda = 1/\sqrt{2} \approx 0.707$** .

Predicted decrease: $m(0) - m(\Delta \mathbf{x}_{\text{nt}}) = \lambda^2/2 = 1/4$. The actual value is $f(-1/2, -1/4) = 0.7315$, a decrease of 0.2685, which is close to the prediction.

(b) Quasi-Newton (4 pts)

Forming the Hessian takes $O(n^2)$ second derivatives, which are often unavailable or expensive. Solving the Newton system then costs $O(n^3)$. BFGS builds a Hessian approximation from gradient differences only and costs $O(n^2)$ per iteration.

Secant condition: $\mathbf{B}_{k+1} \mathbf{s}_k = \mathbf{y}_k$, where $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$.

The curvature condition $\mathbf{s}_k^T \mathbf{y}_k > 0$ (guaranteed by a Wolfe line search) keeps $\mathbf{B}_{k+1}$ positive definite.

Question 7 — Lagrange Multipliers and KKT

15 points

(a) KKT (4 pts, 1 per condition)

Write the constraint as $g(\mathbf{x}, \mathbf{y}) = 5 - \mathbf{x} - 2\mathbf{y} \leq 0$. Lagrangian: $L = \mathbf{x}^2 + \mathbf{y}^2 + \mu(5 - \mathbf{x} - 2\mathbf{y})$.

1. **Stationarity:** $2\mathbf{x} - \mu = 0$, $2\mathbf{y} - 2\mu = 0$

2. **Primal feasibility:** $\mathbf{x} + 2\mathbf{y} \geq 5$

3. **Dual feasibility:** $\mu \geq 0$

4. **Complementary slackness:** $\mu(5 - \mathbf{x} - 2\mathbf{y}) = 0$

(b) Solve (7 pts)

Case $\mu = 0$: stationarity gives $\mathbf{x} = \mathbf{y} = 0$, but then $0 \geq 5$ is false. Infeasible, so no solution in this case.

Case $\mu > 0$: the constraint is active, $\mathbf{x} + 2\mathbf{y} = 5$. Stationarity gives $\mathbf{x} = \mu/2$ and $\mathbf{y} = \mu$, so $\mu/2 + 2\mu = 5$ and $\mu = 2$.

$(\mathbf{x}^*, \mathbf{y}^*) = (1, 2)$, $\mu^* = 2 \geq 0 \checkmark$, $f^* = 5$.

Global: the objective is convex ($\nabla^2 f = 2I > 0$) and the constraint is affine, so the problem is convex and the KKT conditions are sufficient. Slater's condition also holds, e.g. $(5, 5)$ is strictly feasible. Geometrically, $(1, 2)$ is the projection of the origin onto the line $x + 2y = 5$, along its normal $(1, 2)$.

(c) Sensitivity (4 pts)

The multiplier is the rate of change of the optimal value with the right-hand side: $\partial f^*/\partial b = \mu^* = 2$.

Estimate: $f^*(5.2) \approx 5 + 2 \cdot 0.2 = 5.4$.

Exact: the optimum is $(b/5)(1, 2)$, so $f^*(b) = b^2/5$ and $f^*(5.2) = 27.04/5 = 5.408$. The error of 0.008 comes from the curvature of $f^*(b)$, which the first-order estimate ignores.